

Number of casualties report

Dataset: dft-road-casualty-statistics-collision-last-5-years.csv

1. Dataset Overview

Item	Value
Number of Rows	5,000
Number of Columns	44
Missing Values	0
Duplicate Rows	0
Target Column	number_of_casualties
Problem Type	Classification

2. Data Quality

Quality Measure	Result
Overall Quality Score	80 / 100
Missing Values	Excellent
Duplicate Rows	Excellent
Data Types	Excellent
Correlation	Poor
Outliers	Poor

3. Target Balance

The selected target appears to be imbalanced.

Class	Proportion
1	0.8150
2	0.1290
3	0.0382
4	0.0114
5	0.0052
6	0.0006
7	0.0004
9	0.0002

4. Machine Learning Results

Model	ACCURACY	PRECISION	RECALL	F1
Random Forest	0.8080	0.7141	0.8080	0.7236
Logistic Regression	0.8010	0.7070	0.8010	0.7326

Decision Tree	0.6830	0.6848	0.6830	0.6838
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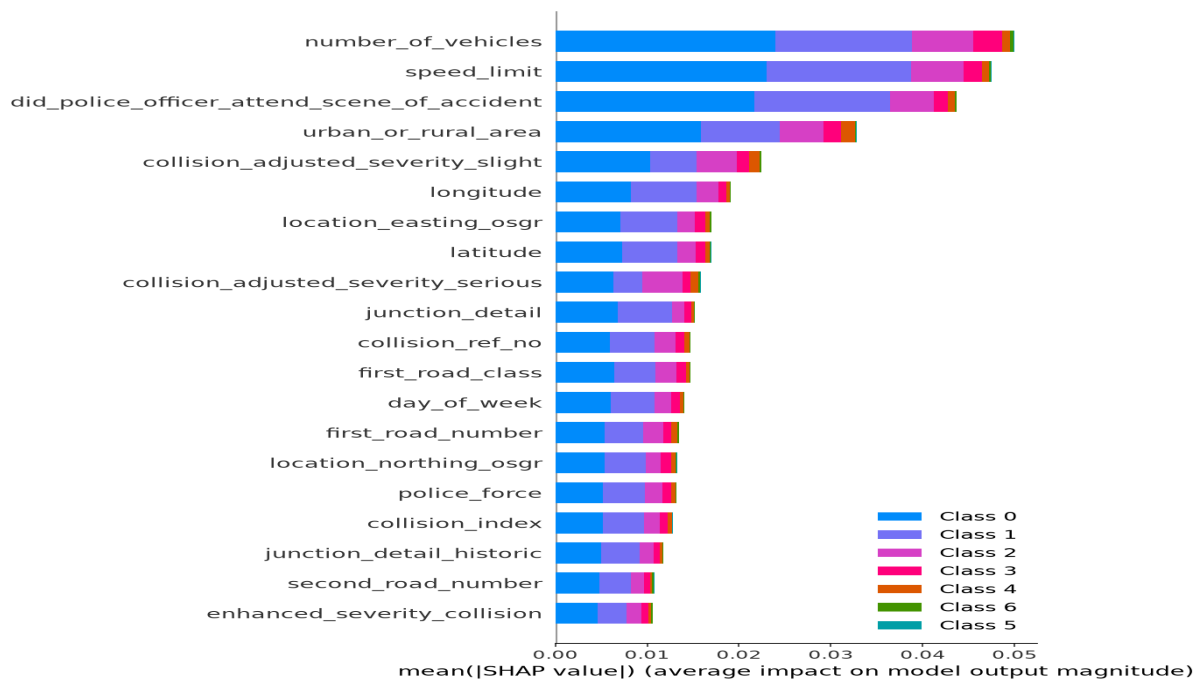
Best Model: Random Forest
ACCURACY: 0.8080

5. Explainable AI / SHAP

SHAP was used to explain which features had the strongest influence on the predictions made by the selected best-performing model.

Rank	Feature	Mean SHAP Value
1	number_of_vehicles	0.0071
2	speed_limit	0.0068
3	did_police_officer_attend_scene_of_accident	0.0062
4	urban_or_rural_area	0.0047
5	collision_adjusted_severity_slight	0.0032
6	longitude	0.0027
7	location_easting_osgr	0.0024
8	latitude	0.0024
9	collision_adjusted_severity_serious	0.0023
10	junction_detail	0.0022

SHAP Summary Visualization



6. Conclusion

The selected target column was 'number_of_casualties'. The task was identified as a classification problem. The best-performing model was Random Forest, with a ACCURACY score of 0.8080. The most influential features identified by SHAP were number_of_vehicles, speed_limit, did_police_officer_attend_scene_of_accident.